



# Biochip Course

## Low-Noise Current Measurement

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a.y. 2022/2023

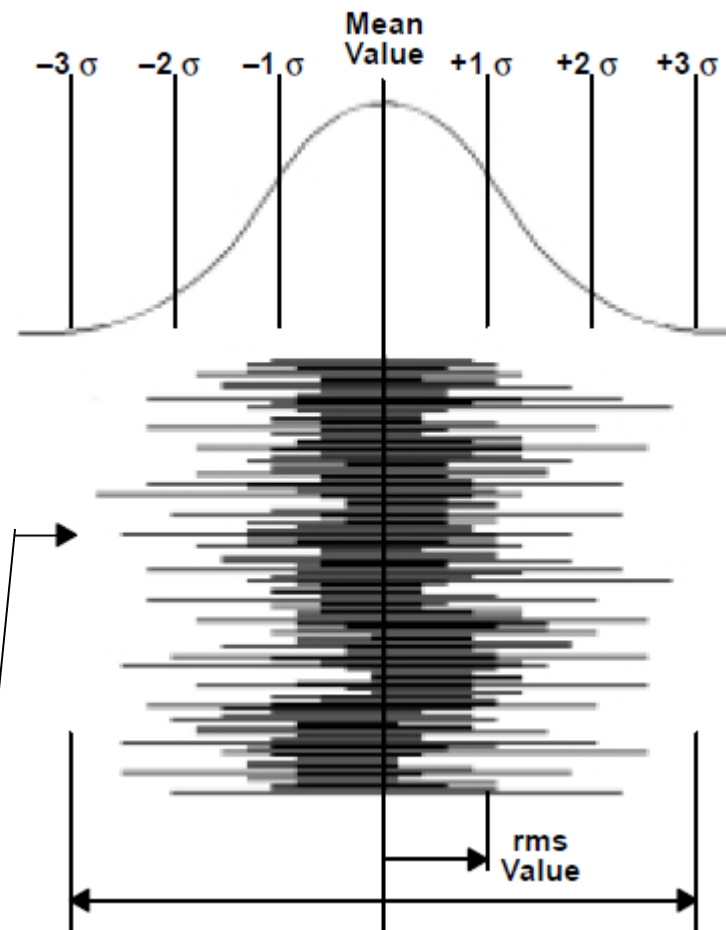
March 22<sup>nd</sup> 2023



# Motivation

Since in electrochemical detection, the current flowing at the working electrode must be accurately measured, here we review the circuits and the techniques adopted for current amplification, in particular with high resolution (i.e. low noise) and for integrated biochips:

$$SNR = \frac{\text{Signal Amplitude}}{\text{Noise Amplitude}} = \frac{\text{Signal}}{\sigma_{rms}}$$



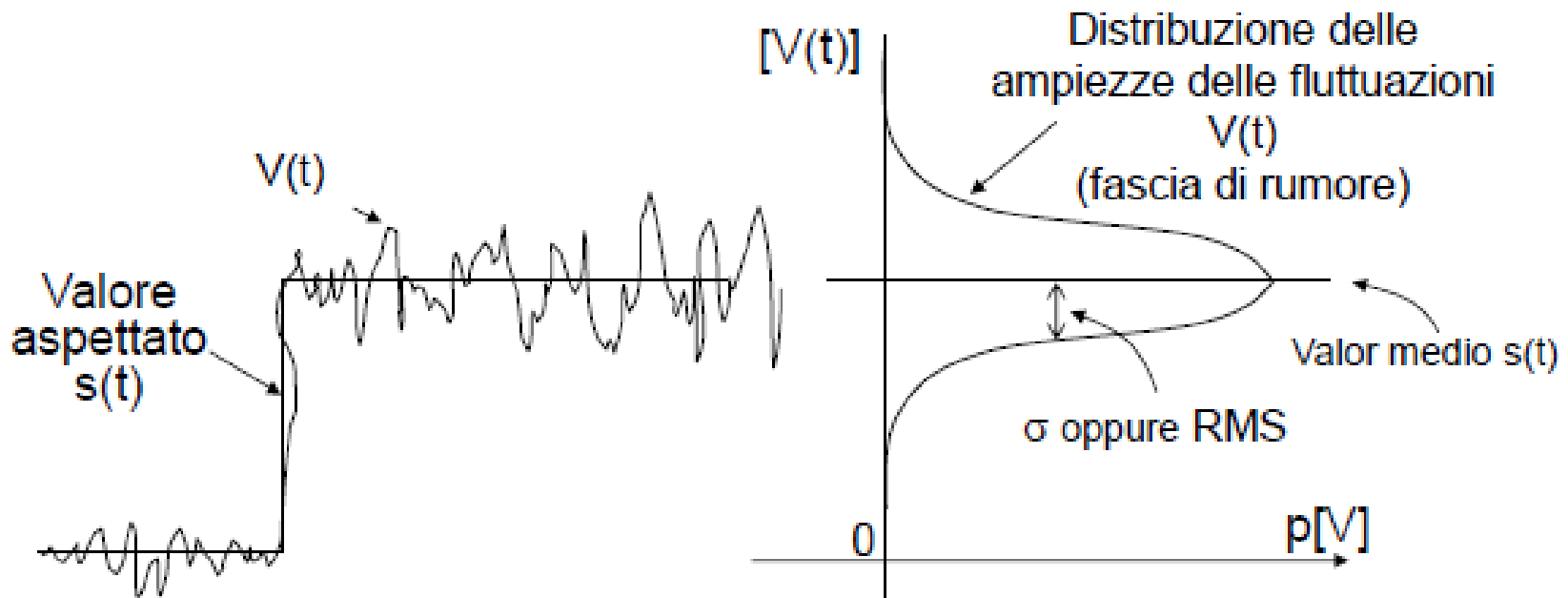


# Fundamentals of Electronic Noise

Any electrical signal is affected by **noise** and disturbances:

$$v(t) = \text{signal}(t) + n(t) + d(t)$$

Noise is a **random fluctuation** of the electrical variables due to the physical behaviour of *internal* components of the circuit

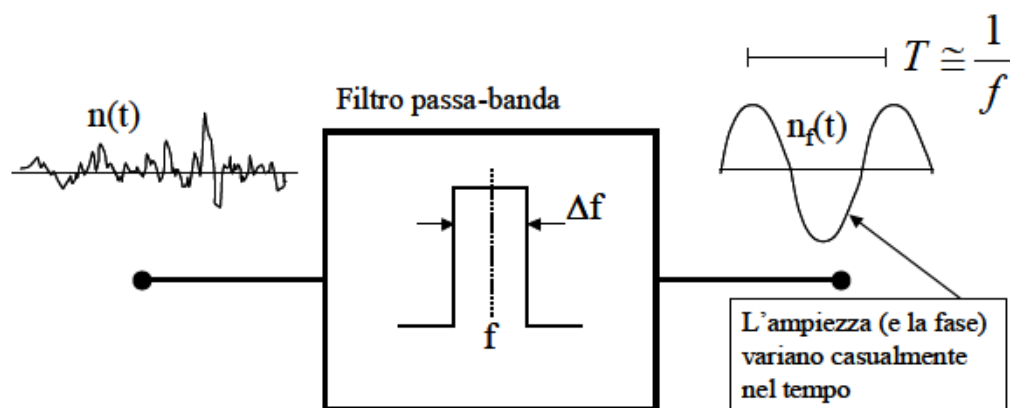


Random process characterized by **standard deviation**  $\sigma = \text{RMS}$



# Noise Power Spectrum

Description in the frequency domain:



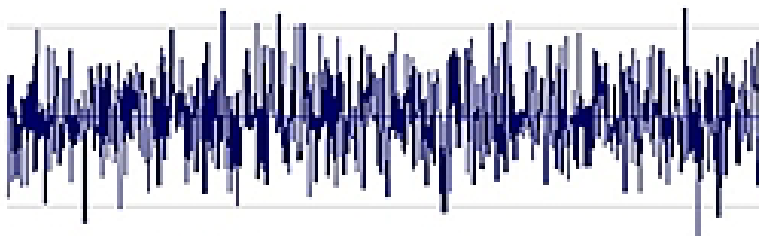
Noise spectral density

$$S(f) = \overline{n_f^2(t)} / \Delta f$$

$$\overline{n^2(t)} = \int_0^\infty S(f) df$$

$$\text{RMS} = \sqrt{\overline{n^2(t)}} = \sqrt{\int_0^\infty S(f) df}$$

Flat-band or  
White Noise



Time Domain Waveform



Spectral Density

1/f flicker  
noise

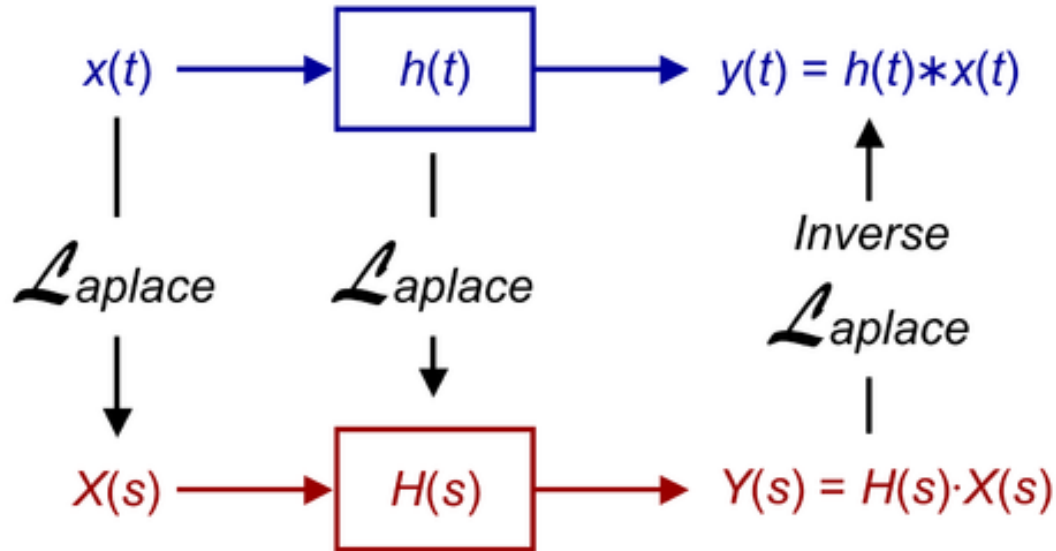




# How to Transfer Noise in the Circuit?

Time domain

Linear system:



Frequency domain

Noise:



Squared magnitude of transfer function (phase has no impact)



# Noise Sources in the Devices: Resistors

Thermal (Johnson) noise: due to the random motion of  $e^-$



- white (flat) power spectrum
- independent of how the resistor is biased
- a  $1k\Omega$  resistor generates  $4nV/\sqrt{\text{Hz}}$  at  $25^\circ\text{C}$



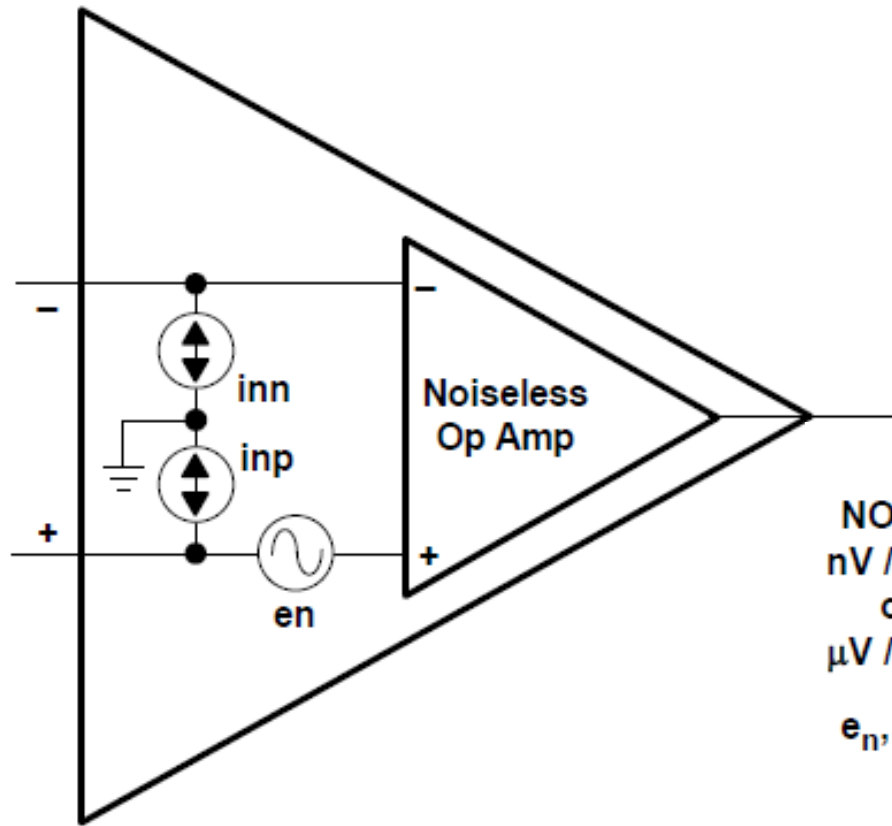
Due to the random arrival times of discrete charge carriers crossing a potential barrier or a depleted region (PN junction)

$$S_{I_c}(f) = 2qI_c \quad [A^2/Hz]$$

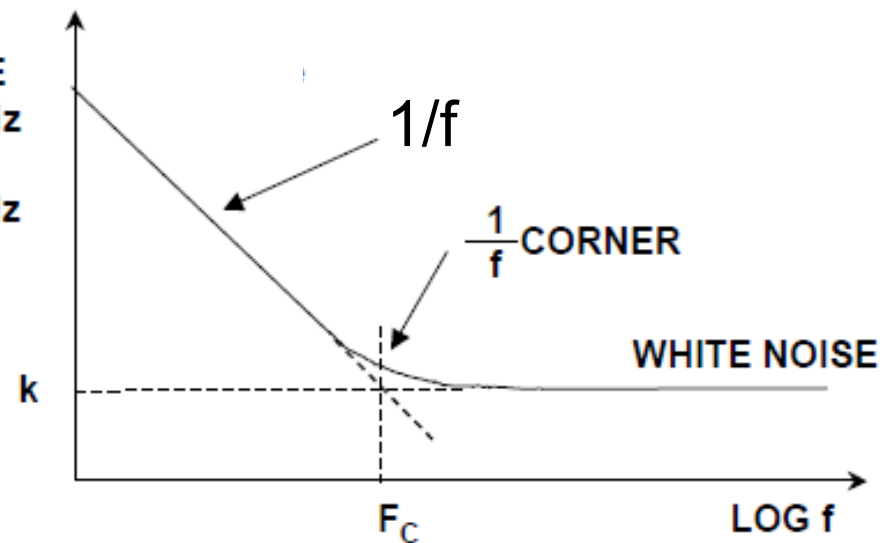
- white (flat) power spectrum
- power density proportional to DC current
- independent of temperature

# Noise in Operational Amplifiers

- Input equivalent noise generators:
- $V_n$  ( $e_n$ ) Voltage noise (in gnd): 2-20nV/ $\sqrt{\text{Hz}}$
  - $I_n$  Current noise (in float): depends on technology



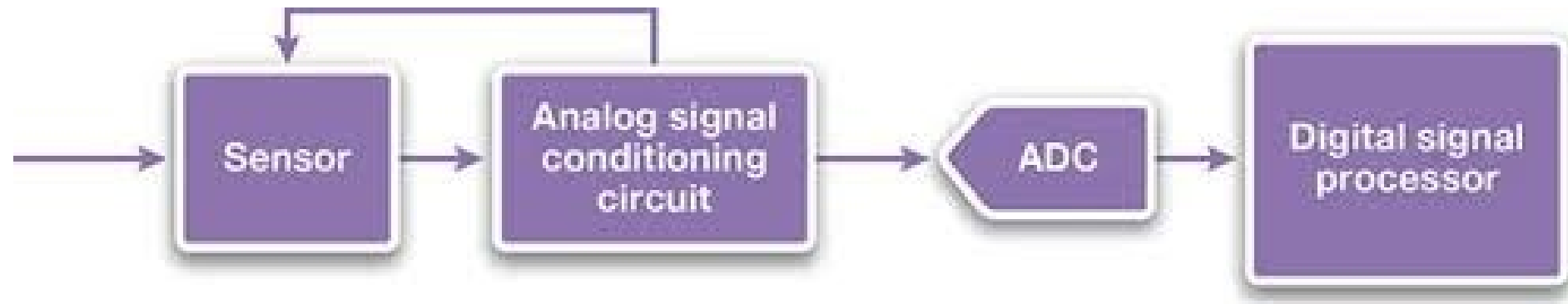
NOISE  
nV /  $\sqrt{\text{Hz}}$   
or  
 $\mu\text{V} / \sqrt{\text{Hz}}$   
 $e_n, i_n$







# Techniques for Measuring a Current



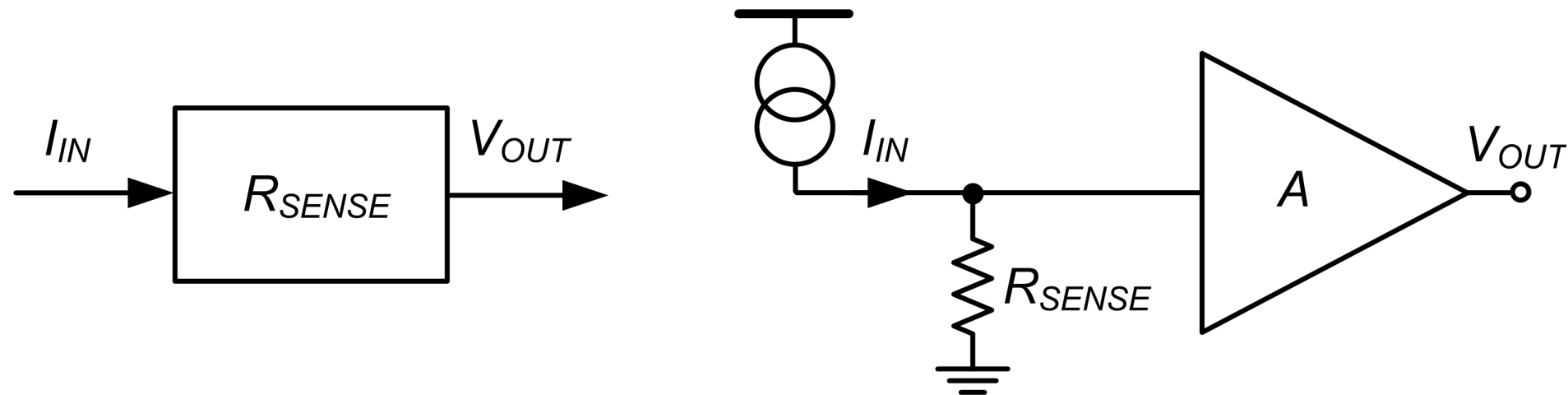
## Tasks of the current conditioning chain:

- Necessary to convert the input current into a voltage for ADC
- Adapt the range to the ADC FSR (gain)
- Improve the SNR (filter the noise)

(Some operations (such as sum) can be performed more easily among currents rather than voltages)

# Passive Current Conversion

First approach: the current is forced to flow in a known resistor  $R_{SENSE}$  to convert it into a voltage (Ohm's law):



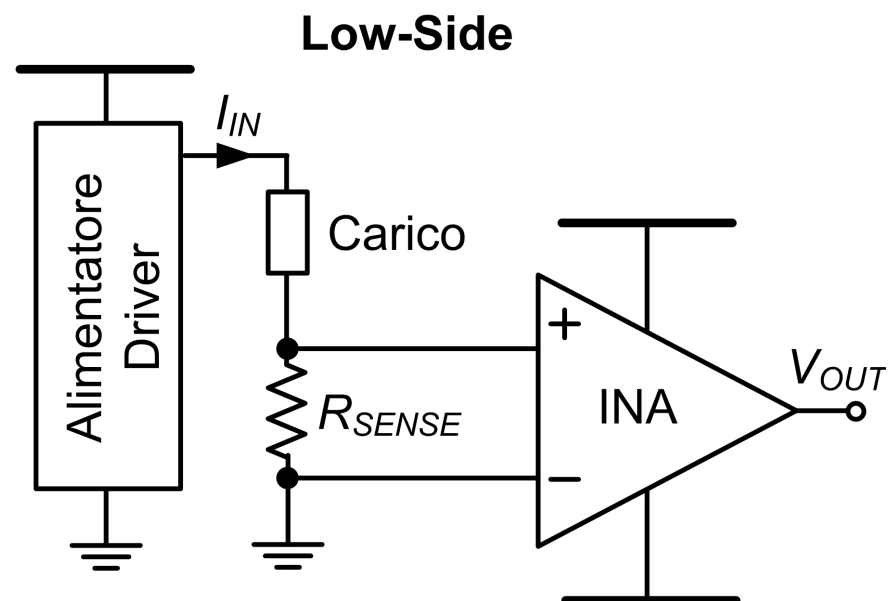
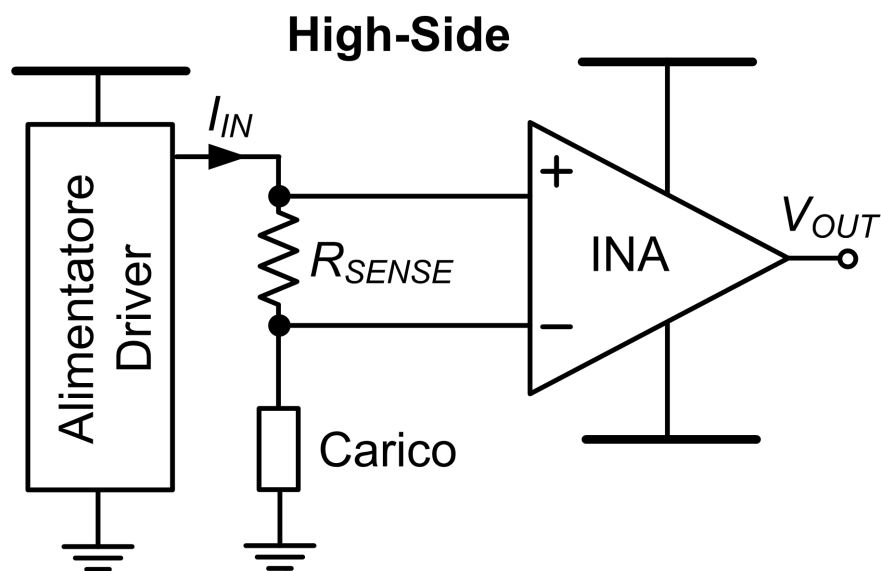
## Advantages:

- passive
- simple
- linear



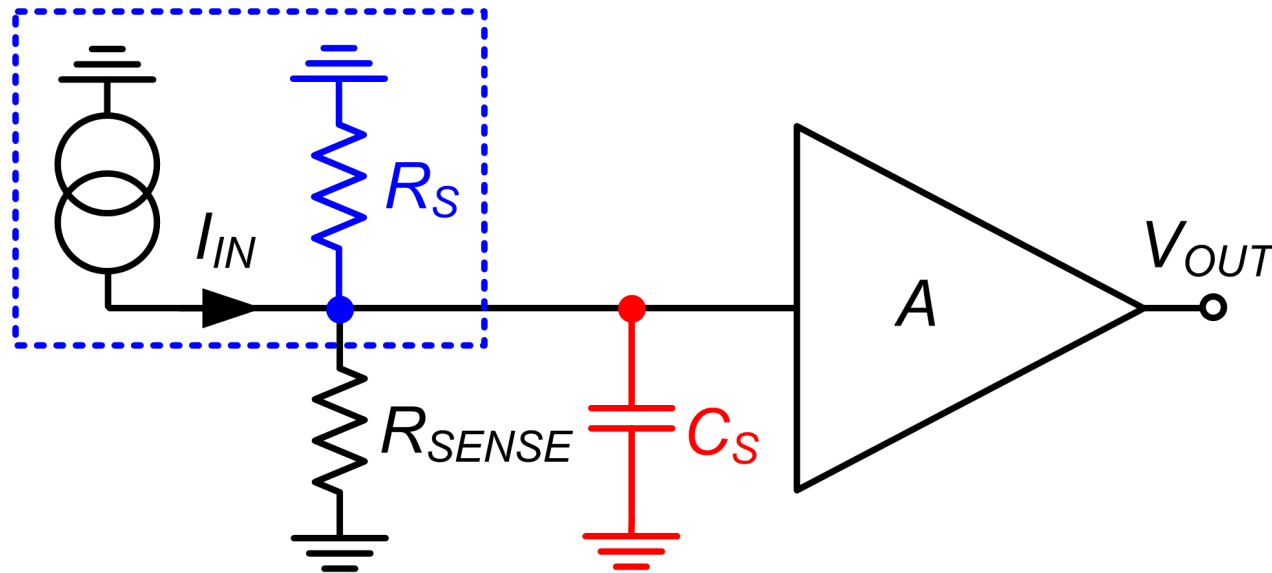
# Current Monitor

Adopted to sense the current with  $R_{SENSE}$  in series to the load:



Trade-off for the value of  $R_{SENSE}$  : conversion gain vs. voltage drop

# Limits of the Passive Conversion



- high input resistance  $R_{SENSE}$  (partition with  $R_S$ )
- sensitive to stray capacitance  $C_S$  (limits the bandwidth)

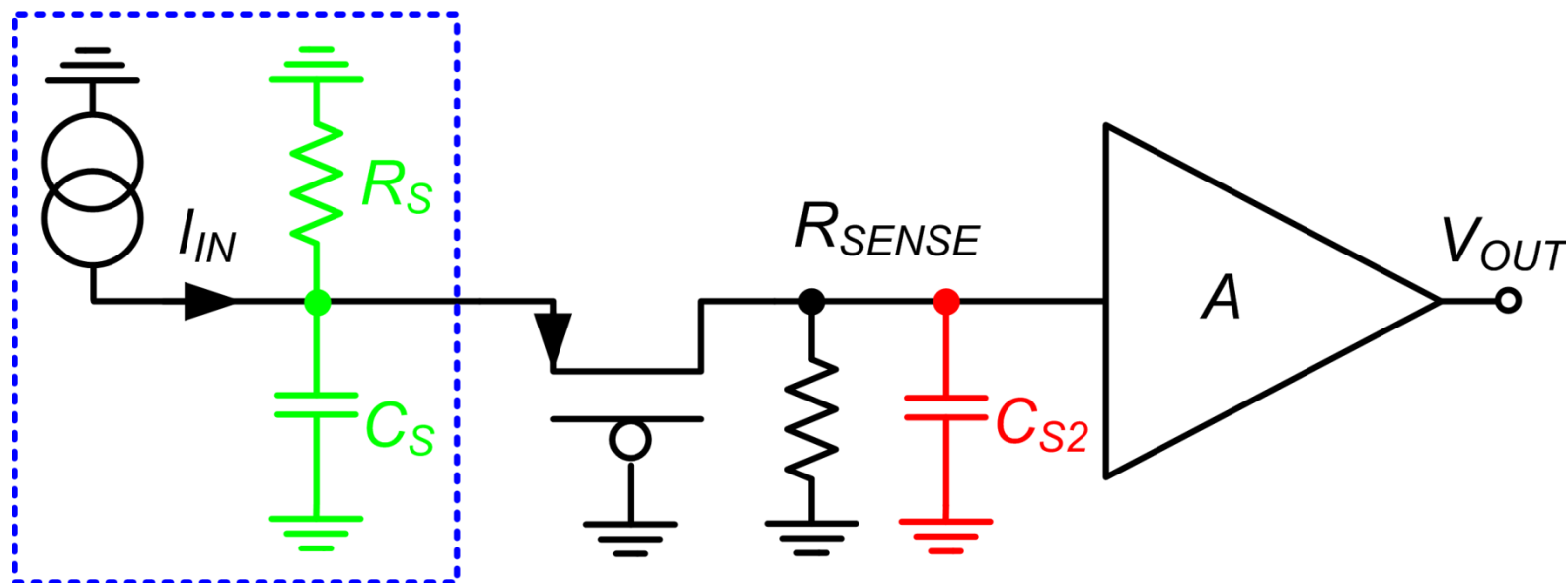


both limits are softened by reducing  $R_{SENSE}$  but thus the gain is reduced!



# Active Current Buffer

An active current buffer can be used to decouple the input impedance and  $R_{SENSE}$  (cascode):

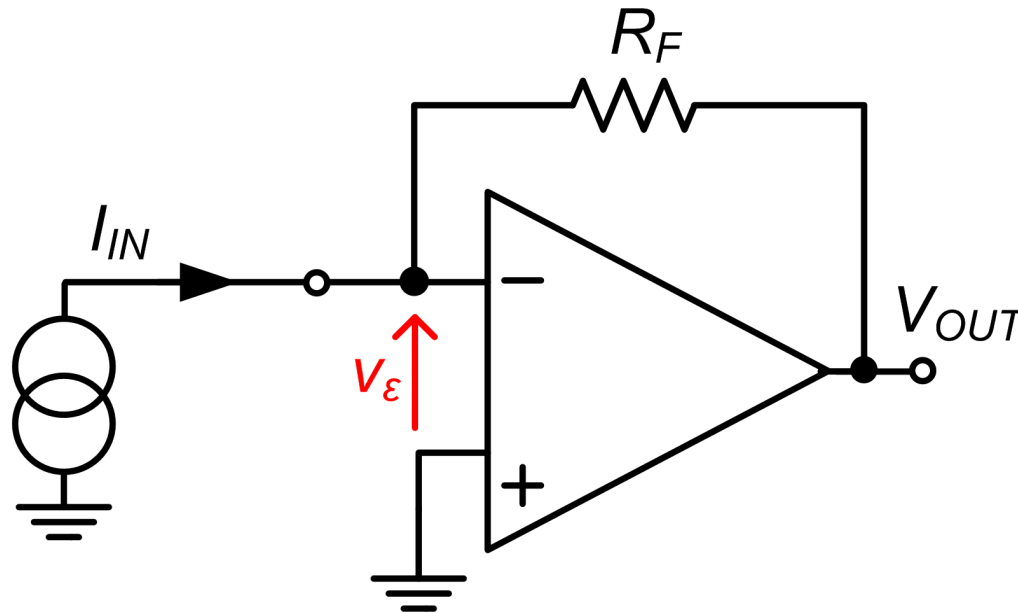


Issues, since the input current is the bias current of the transistor:

- high input impedance (for small currents)
- monodirectional

# The Transimpedance Amplifier

Active conversion based on negative feedback:

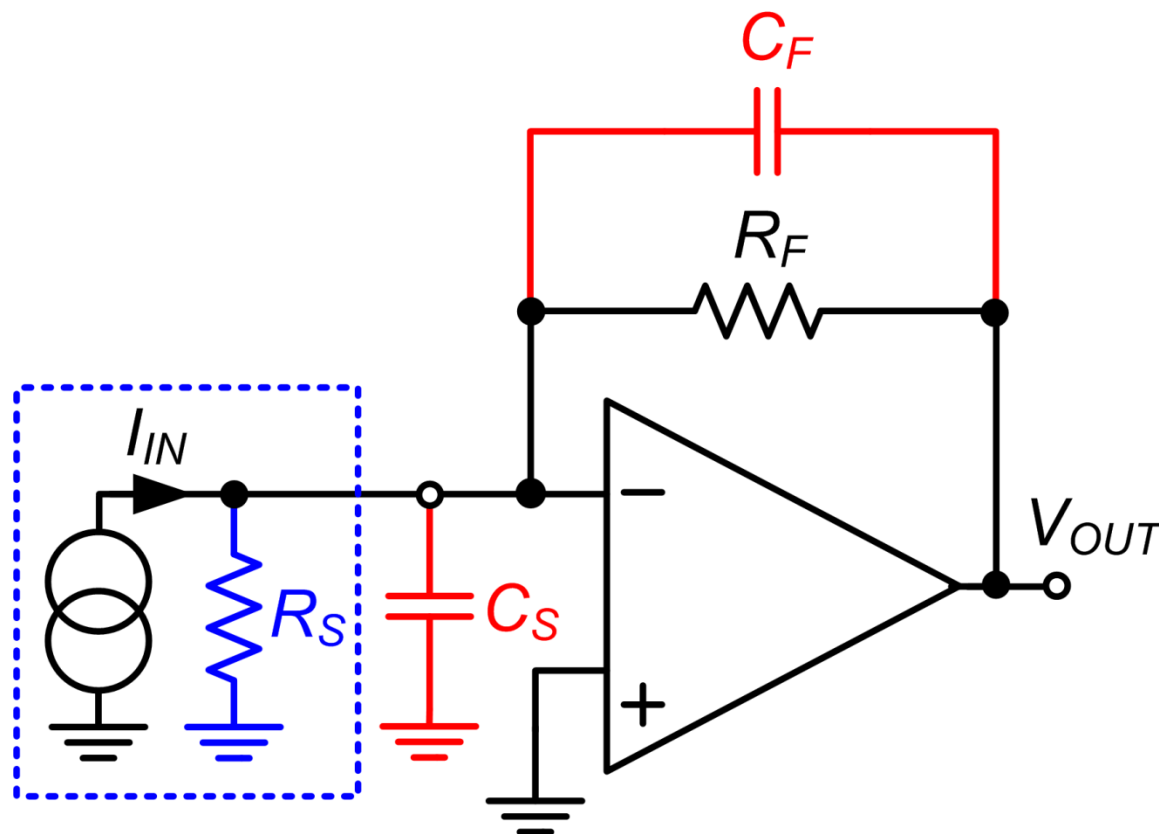


The feedback tends to cancel the control voltage ( $V_{error}$ ) keeping the input at *virtual ground*, the whole current must flow in  $R_F$

$$V_{OUT} = -I_{IN} \cdot R_F$$



# Advantages of the Transimpedance Stage



To pay attention:

- stability
- noise
- dynamic range

Thanks to the feedback, the impact of  $C_S$  and  $R_S$  is cancelled!

The only dangerous parasitic is  $C_F$  (0.2pF for a SMT package)

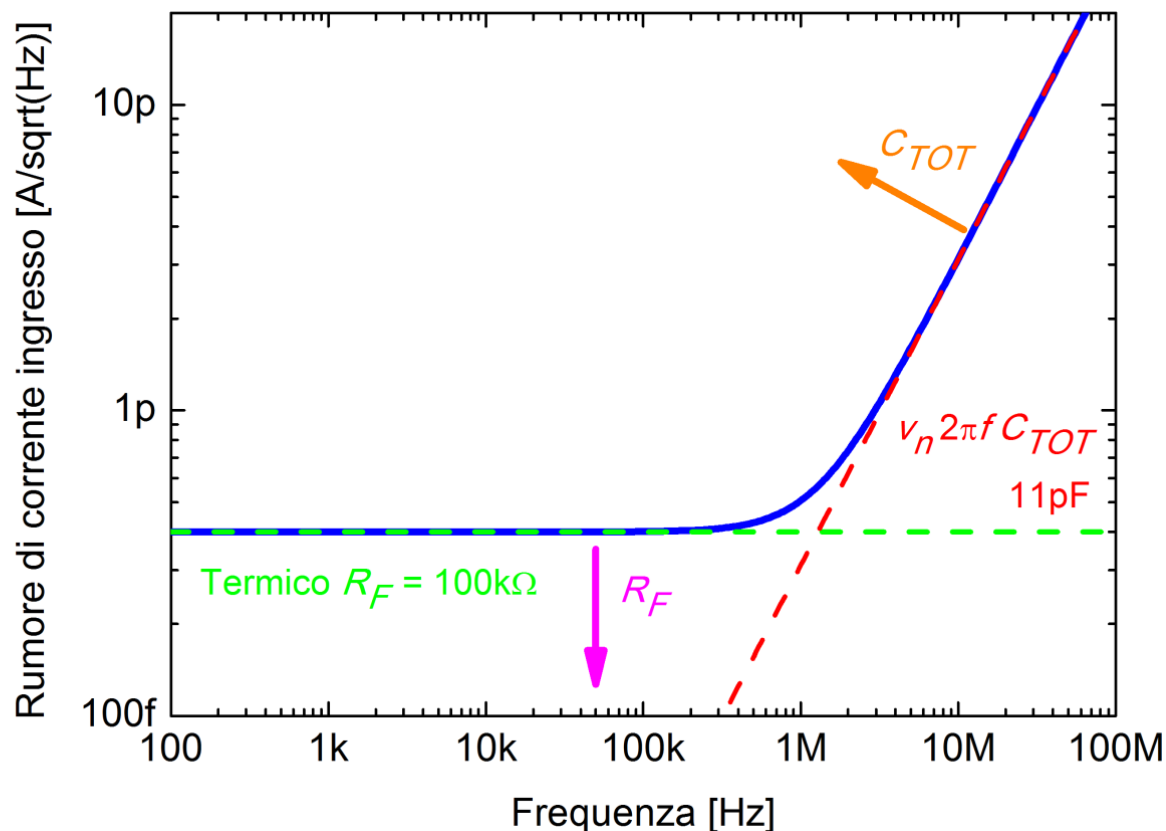
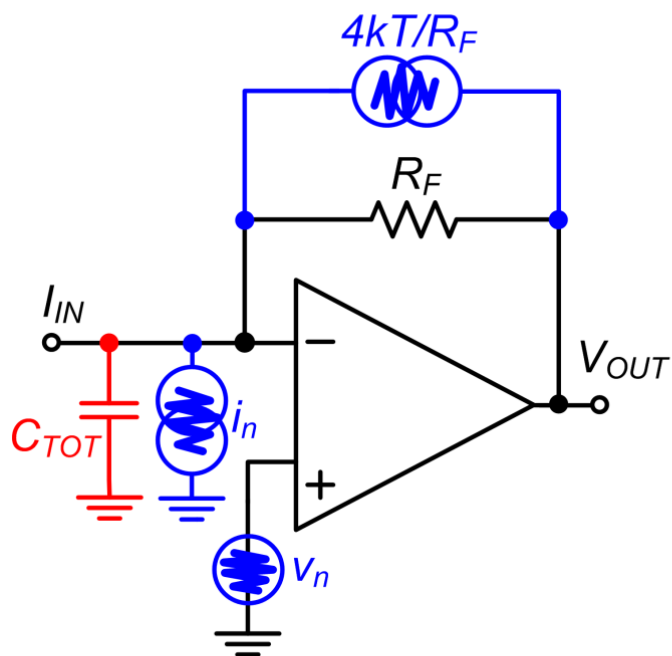


# Noise Analysis

Input-referred total current noise:

$$S_i(f) = \sqrt{i_n^2 + \frac{4kT}{R_F} + \left(\frac{v_n}{R_F}\right)^2 + (v_n 2\pi f C_{TOT})^2}$$

Minimize  $C_S$ !







# Limits of the Transimpedance Amplifier

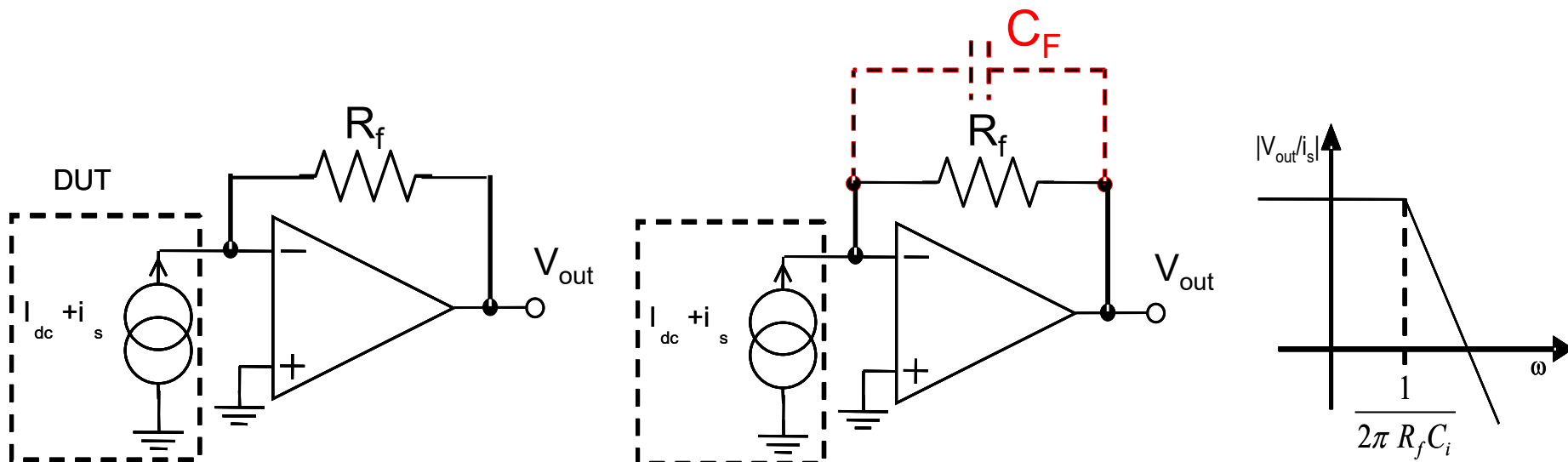
A single component sets all the properties (contrasting trends):

- the gain  $\sim R_F$
- the noise  $\sim \sqrt{4kT/R_F}$
- the bandwidth  $\sim 1/(R_F C_F)$



**Resolution/speed trade-off**

For example:  $R_F = 1\text{G}\Omega$   $C_F = 0.5\text{pF} \rightarrow \text{BW} = 300\text{Hz}$   $S_i = 5\text{fA}/\sqrt{\text{Hz}}$





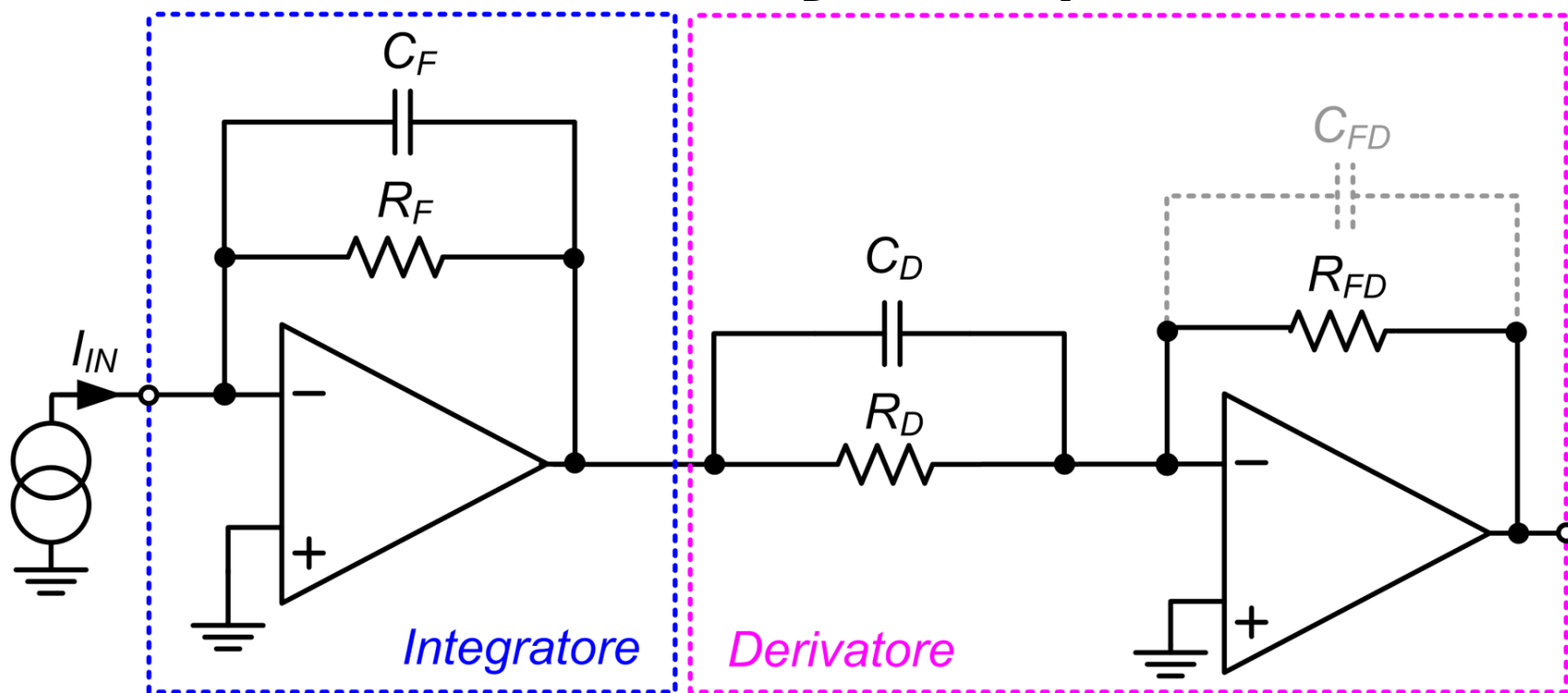
# **Advanced topologies allowing relaxing this trade-off**



# Pseudo Integrator-Differentiator Cascade

Introduce a zero that cancels the pole:  $R_F \cdot C_F = R_D \cdot C_D$

$$V_{OUT} = R_{FD} \frac{R_F}{R_D} = R_{FD} \frac{C_D}{C_F}$$

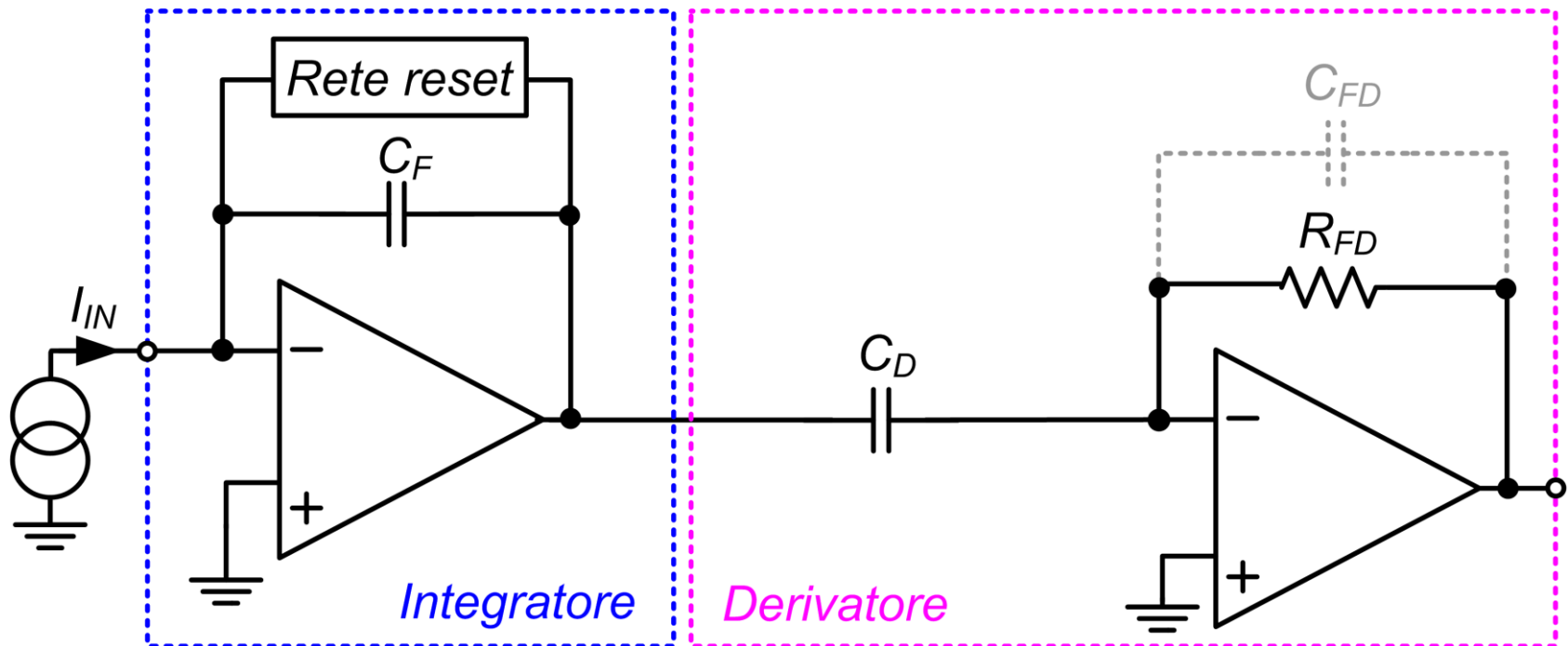


Pro: extends the bandwidth, preserving the lower noise

Con: manual tuning of the zero (due to parasitics)

# Integrator-Differentiator Cascade

$R_F$  is the noisy component and is removed from the integrator  
However, due to input DC current components (input signal, opamp bias), a reset network is necessary





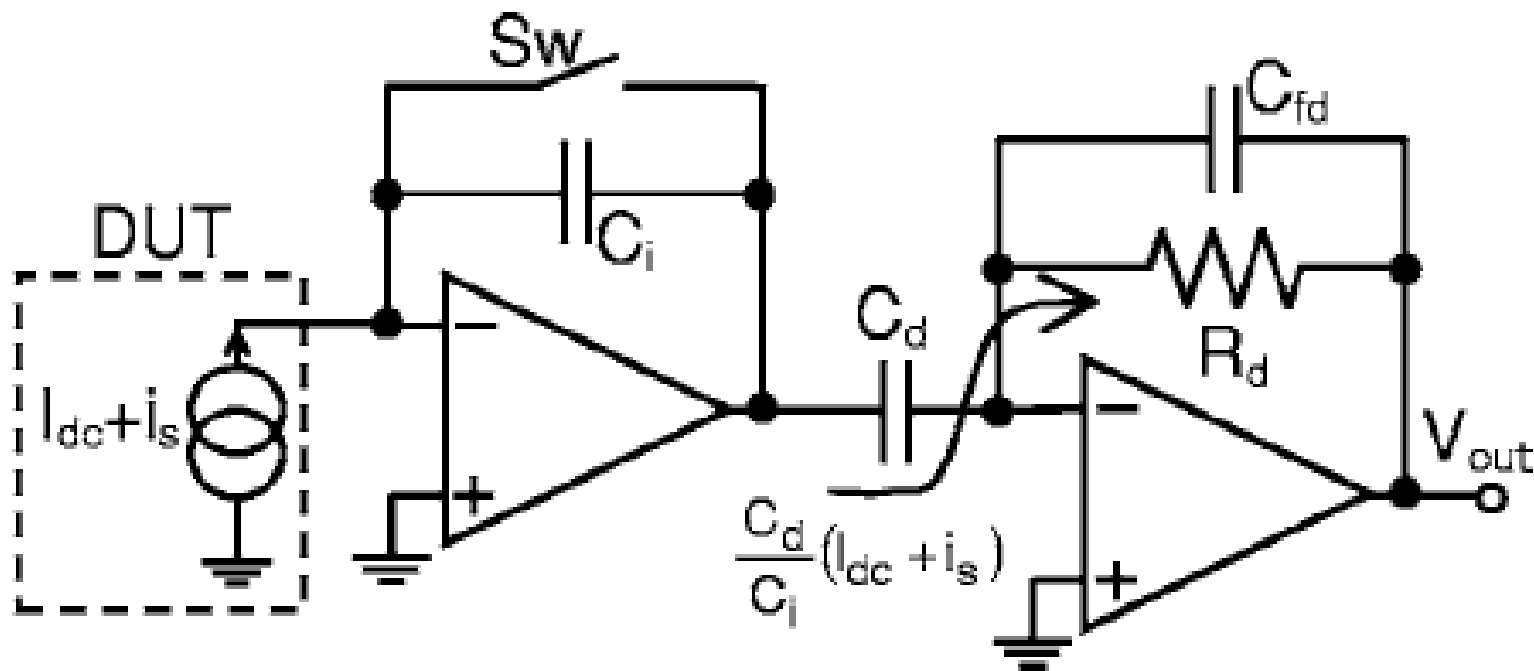
# Switch Reset

Switch periodically closed to discharge the capacitor:



Limits the maximum measurement time

For example:  $I_{IN-DC} = 1\text{nA}$ ,  $C_F = 1\text{pF} \rightarrow (5\text{V}) T_{MAX} = 5\text{ms}$

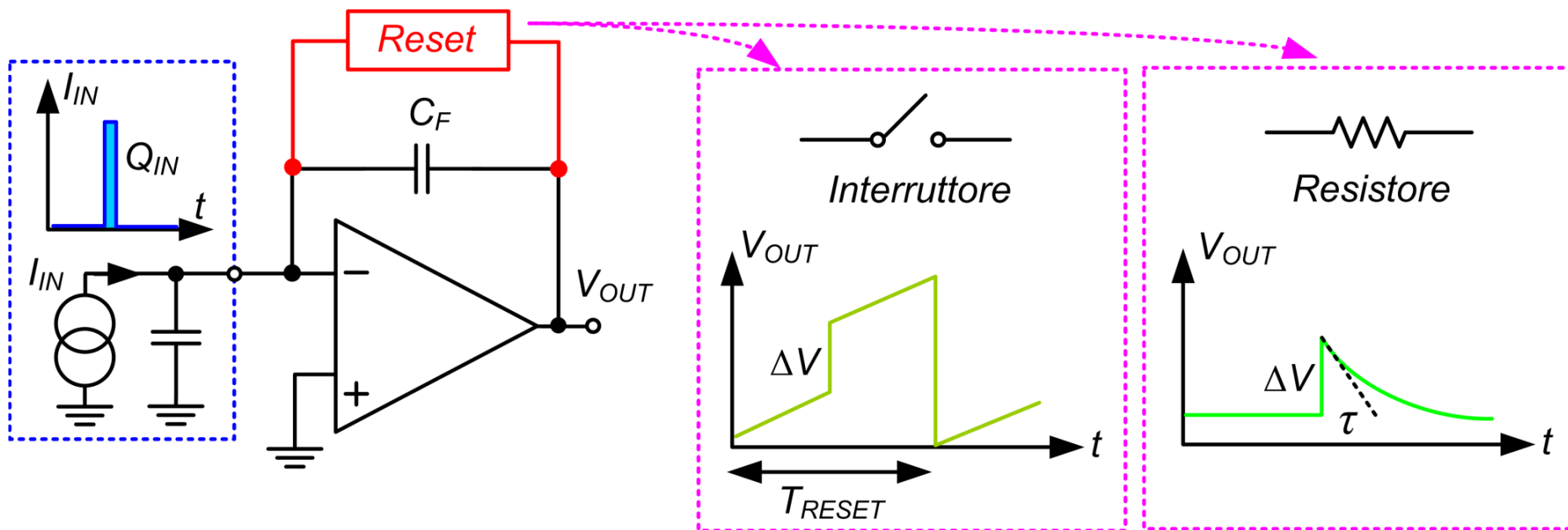




# The Charge Amplifier

Particular case of current sensing: integral of the current is the collected charge (for piezoelectric sensors):

$$V_{OUT} = -\frac{1}{C_F} \int i_{IN}(t) dt = -\frac{Q_{IN}}{C_F}$$

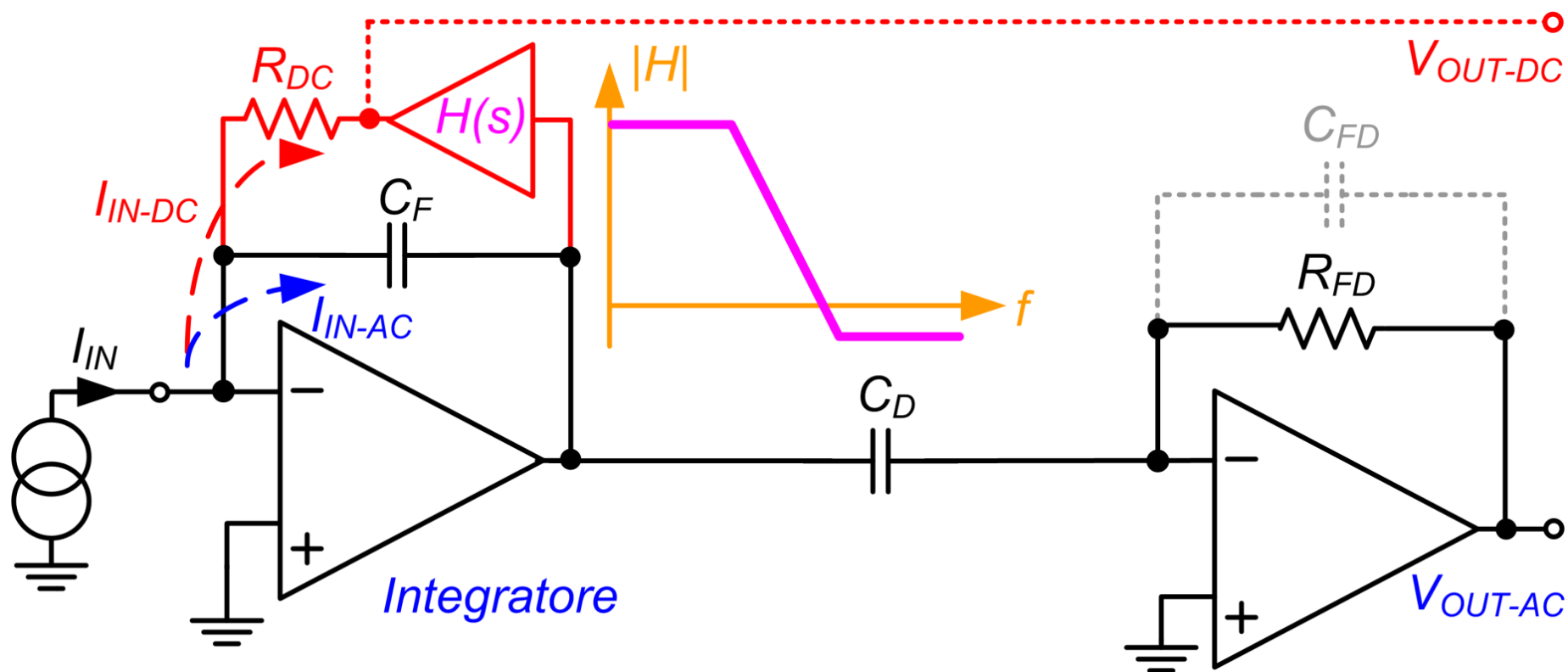


Continuous or gated reset.



# Active Reset

To achieve continuous-time operation: additional feedback branch  $H(s)$  with high gain at low frequency:





# Transfer Functions

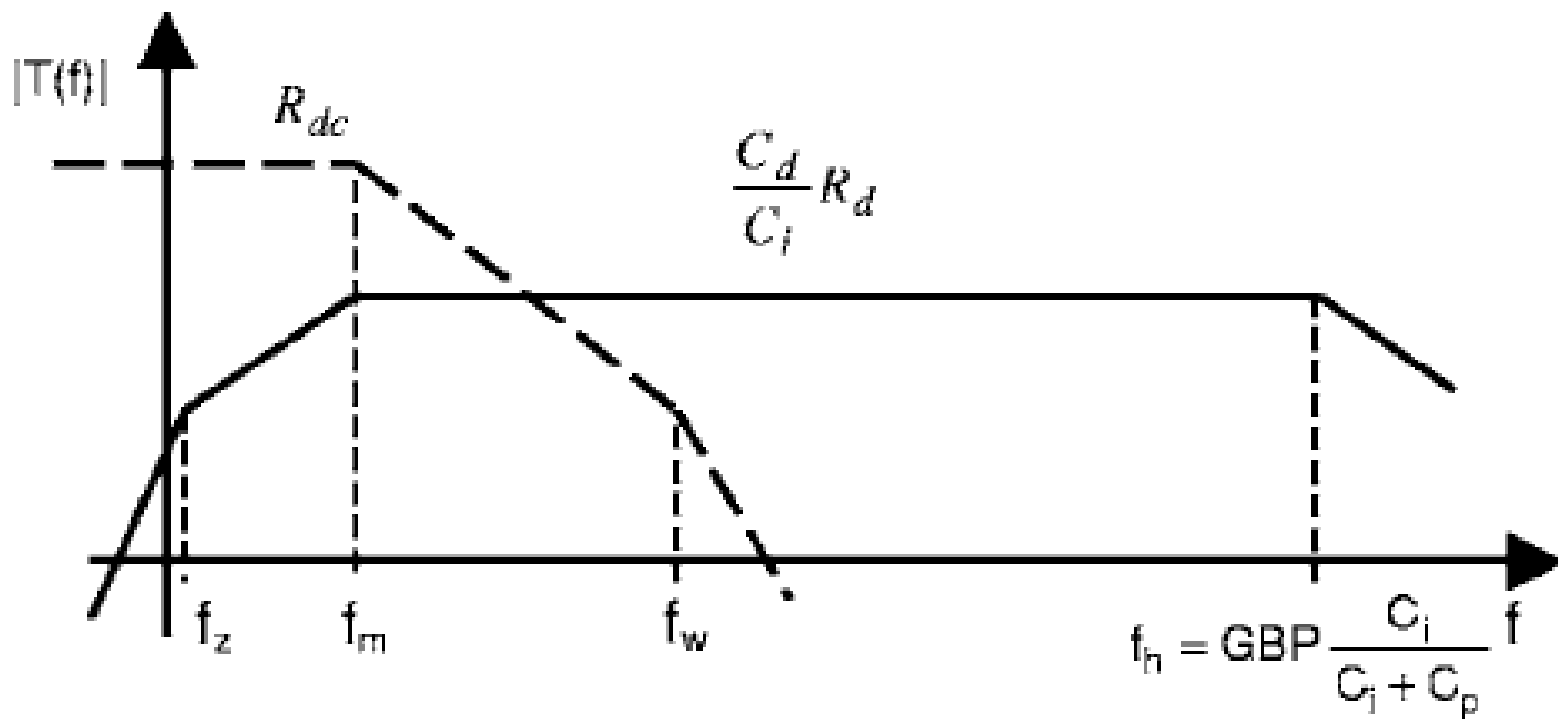
Two separate paths and gains:

- DC and low frequency

$$V_{OUT-DC} = -R_{DC}I_{IN}$$

- High frequency

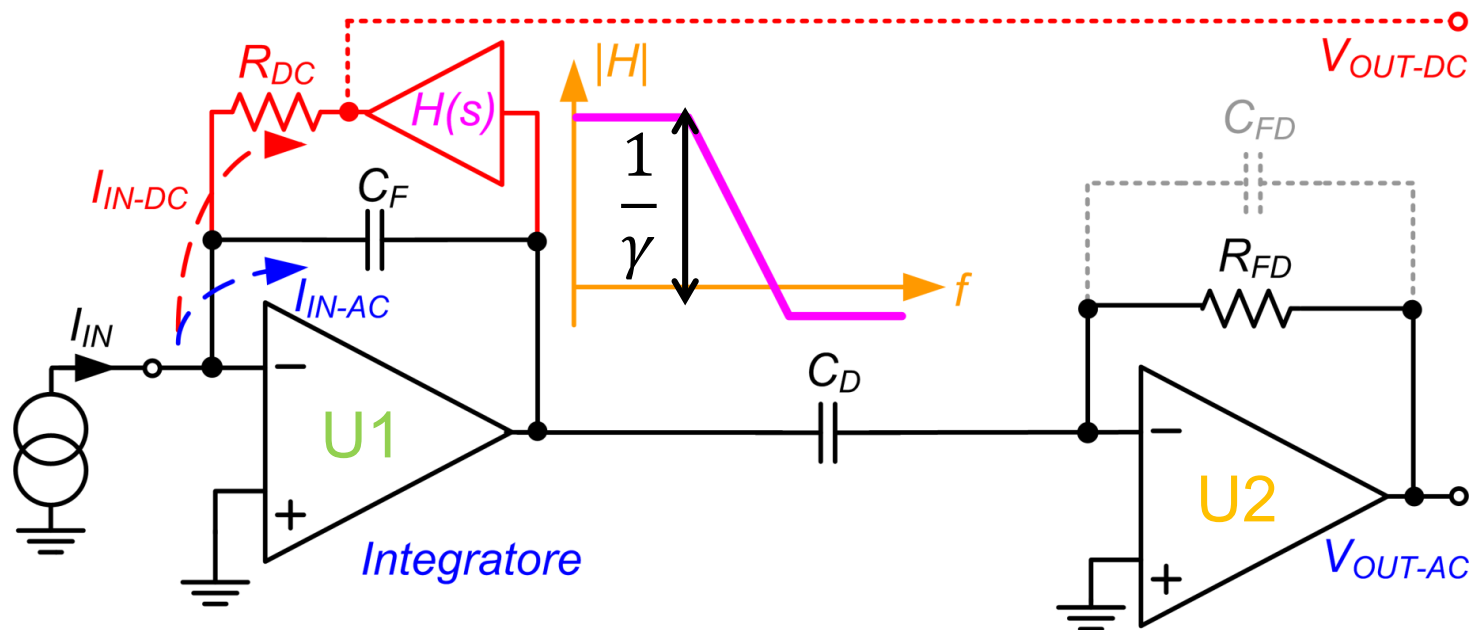
$$V_{OUT-AC} = \frac{C_D}{C_F} R_{FD} I_{IN}$$







# Noise Analysis

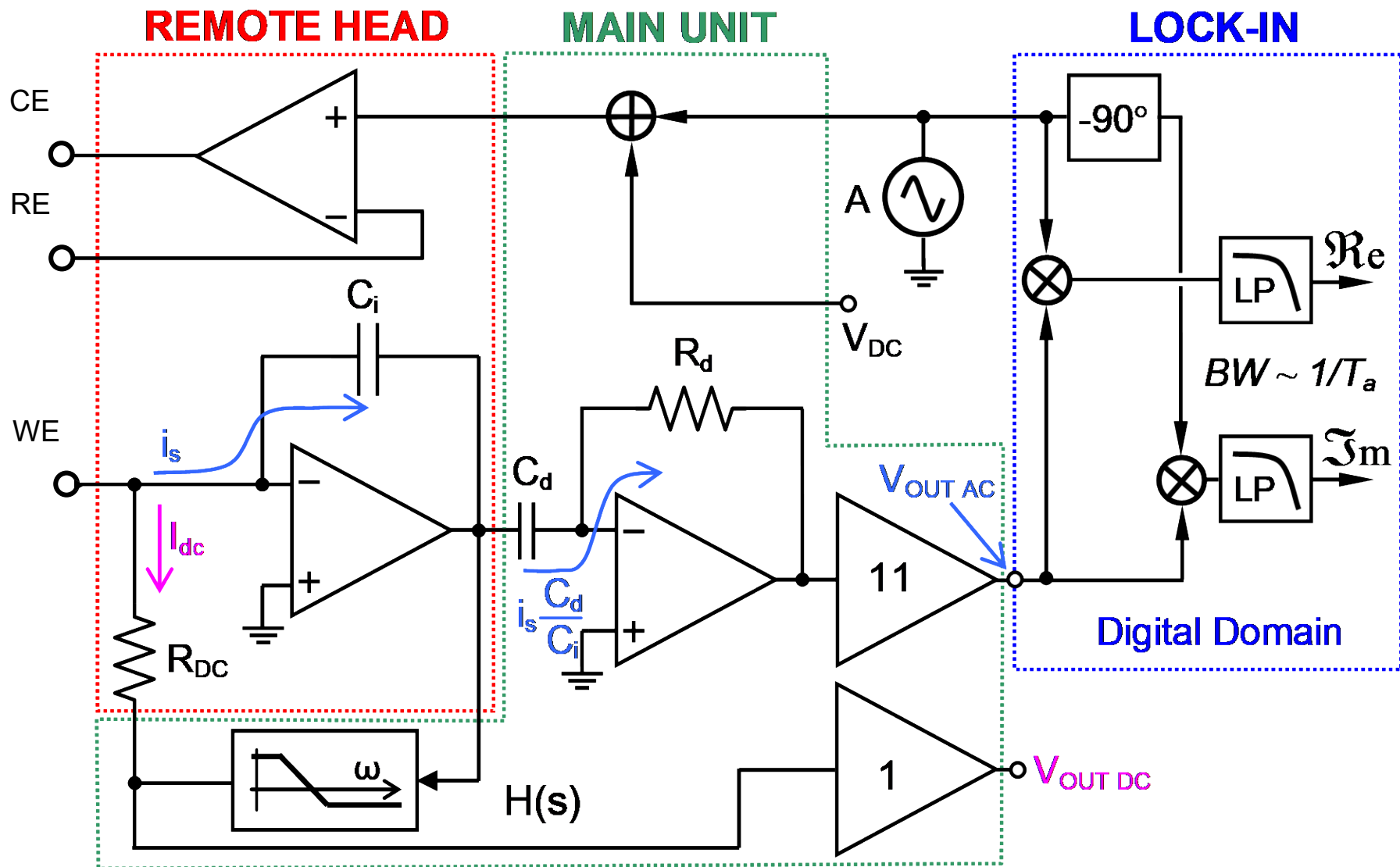


Attenuation of U2 noise, but additional noise of  $R_{DC}-H(S)$

$$S_i(f) = \sqrt{\underbrace{i_1^2 + \left(\frac{v_1}{R_{DC}}\right)^2 + (v_1 2\pi f C_F)^2}_{U1} + \underbrace{\left(i_2^2 + \left(\frac{v_2}{R_{FD}}\right)^2 + (v_2 2\pi f C_D)^2 + \frac{4kT}{R_{FD}}\right)\left(\frac{C_F}{C_D}\right)^2}_{U2} + \underbrace{\frac{4kT}{R_{DC}} + \left(\frac{v_3}{\gamma R_{DC}}\right)^2}_{H(s)}}$$

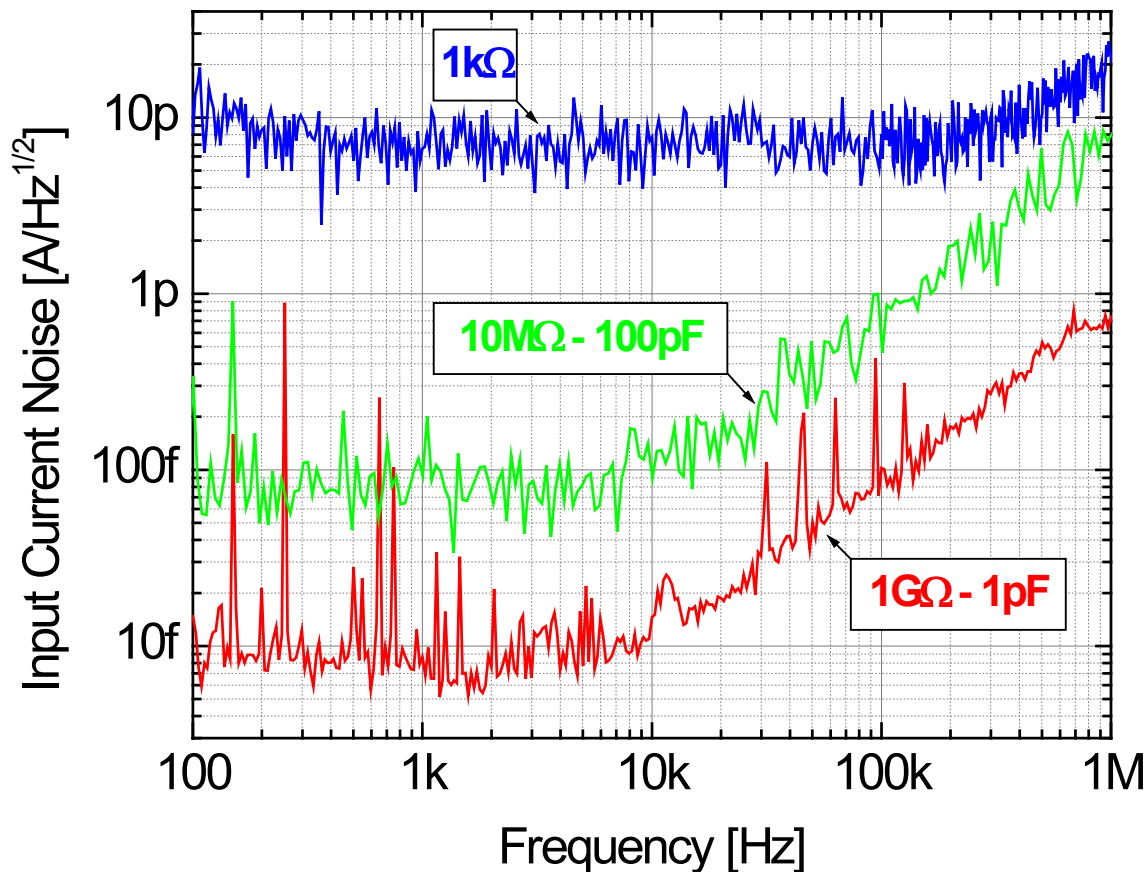


# Enhanced Potentiostat Architecture





# Experimental Results



$I_{\text{DC}}$  max 4mA

$I_{\text{DC}}$  max 1 $\mu\text{A}$

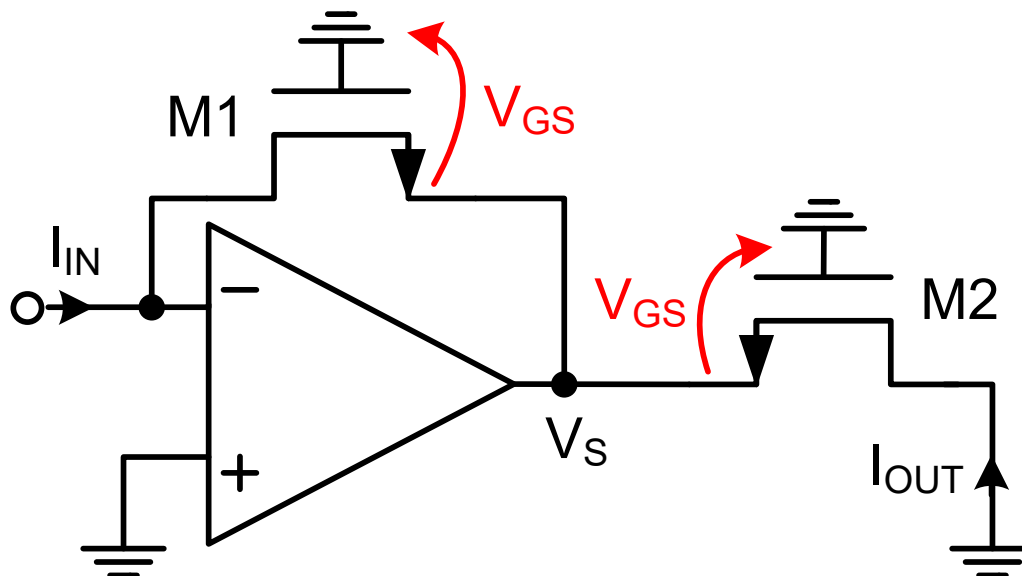
$I_{\text{DC}}$  max 10nA

| HEAD                                                                 | $R_{\text{DC}}$ | $C_i$ | $G_{\text{DC}}$ | $G_{\text{AC}}$ | BW    | Noise         | Max. $I_{\text{DC}}$ |
|----------------------------------------------------------------------|-----------------|-------|-----------------|-----------------|-------|---------------|----------------------|
| CMOS                                                                 | 45M $\Omega$    | 0.1pF | 45M $\Omega$    | 66M $\Omega$    | 2MHz  | 3fA/ sqrt(Hz) | 10nA                 |
| Integrator<br>different.                                             | 1G $\Omega$     | 1pF   | 1G $\Omega$     | 400M $\Omega$   | 1MHz  | 7fA/sqrt(Hz)  | 10nA                 |
|                                                                      | 1G $\Omega$     | 10pF  | 1G $\Omega$     | 40M $\Omega$    | 1MHz  | 10fA/sqrt(Hz) | 10nA                 |
|                                                                      | 10M $\Omega$    | 100pF | 10M $\Omega$    | 4M $\Omega$     | 1MHz  | 70fA/sqrt(Hz) | 1 $\mu\text{A}$      |
| standard transimpedance with $R_{\text{feedback}} = 1\text{k}\Omega$ |                 |       |                 |                 | 10MHz | 6pA/sqrt(Hz)  | 4mA                  |



# Matched-MOSFETs Current Amplifier

Both transistors operate in the same condition

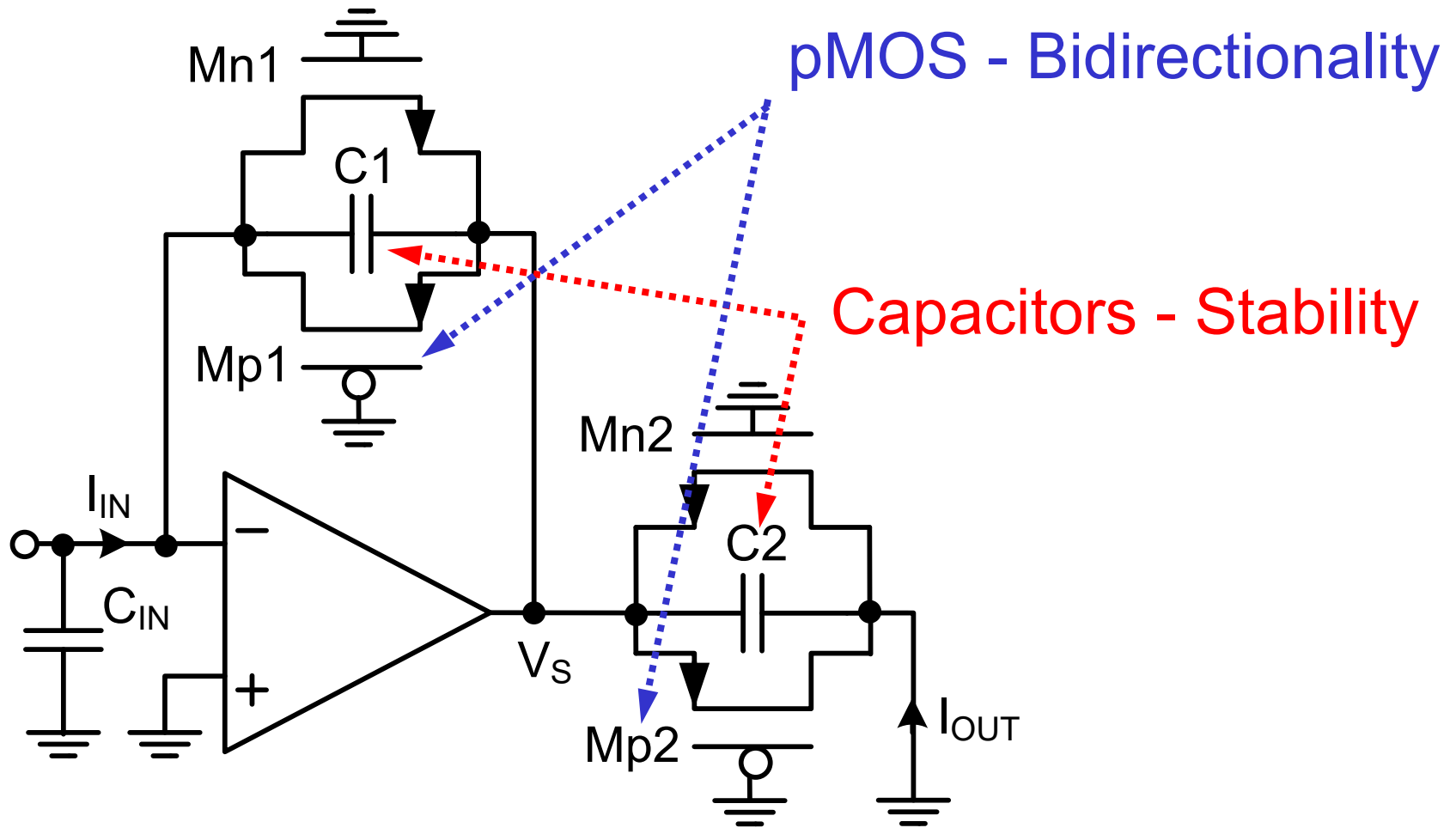


$$\text{GAIN} = \frac{I_{\text{OUT}}}{I_{\text{IN}}} = \frac{I_{\text{M2}}}{I_{\text{M1}}} = \frac{W_2}{W_1} \cdot \frac{L_1}{L_2}$$

M2 is implemented as the parallel of N replicas of M1

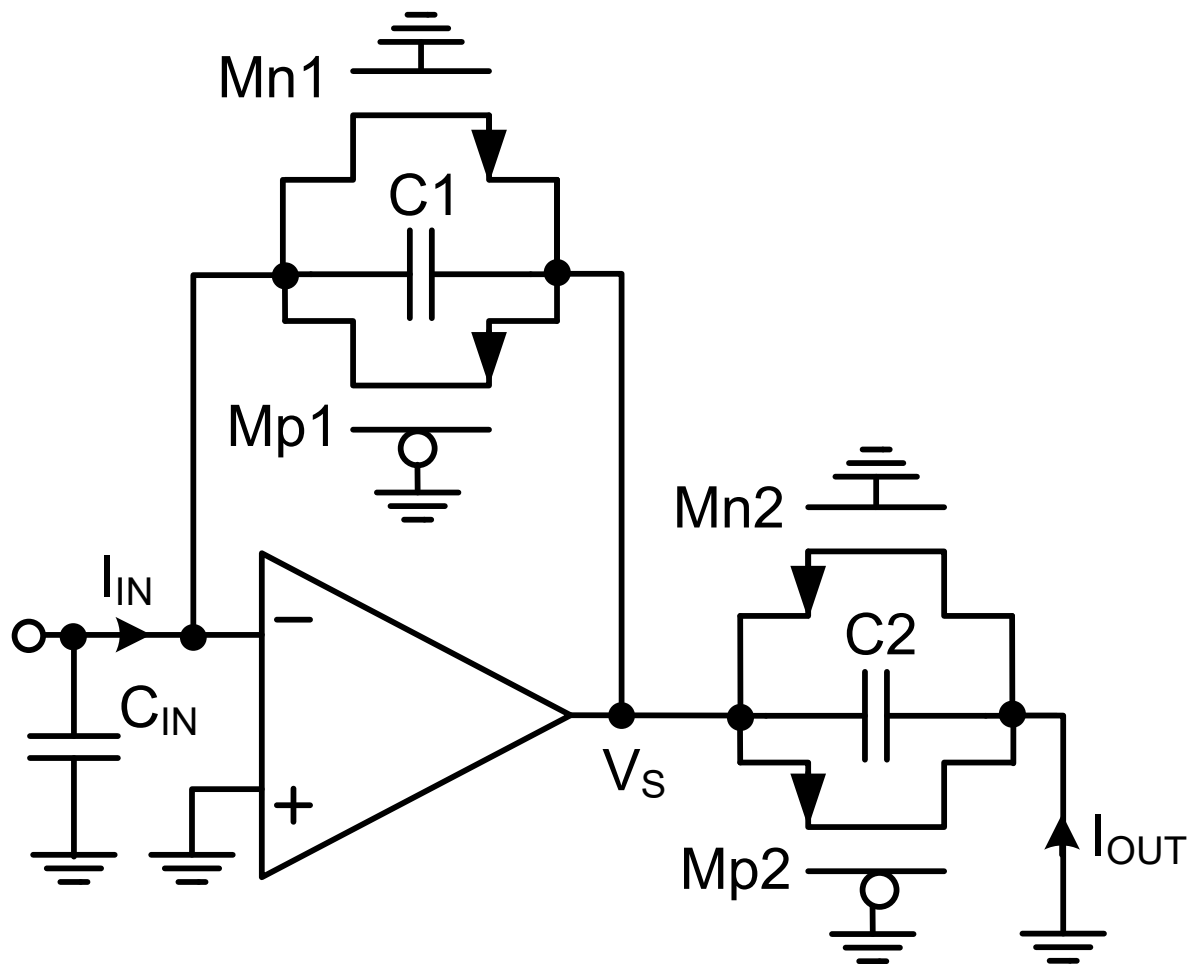


# Improved Configuration





# Noise Analysis

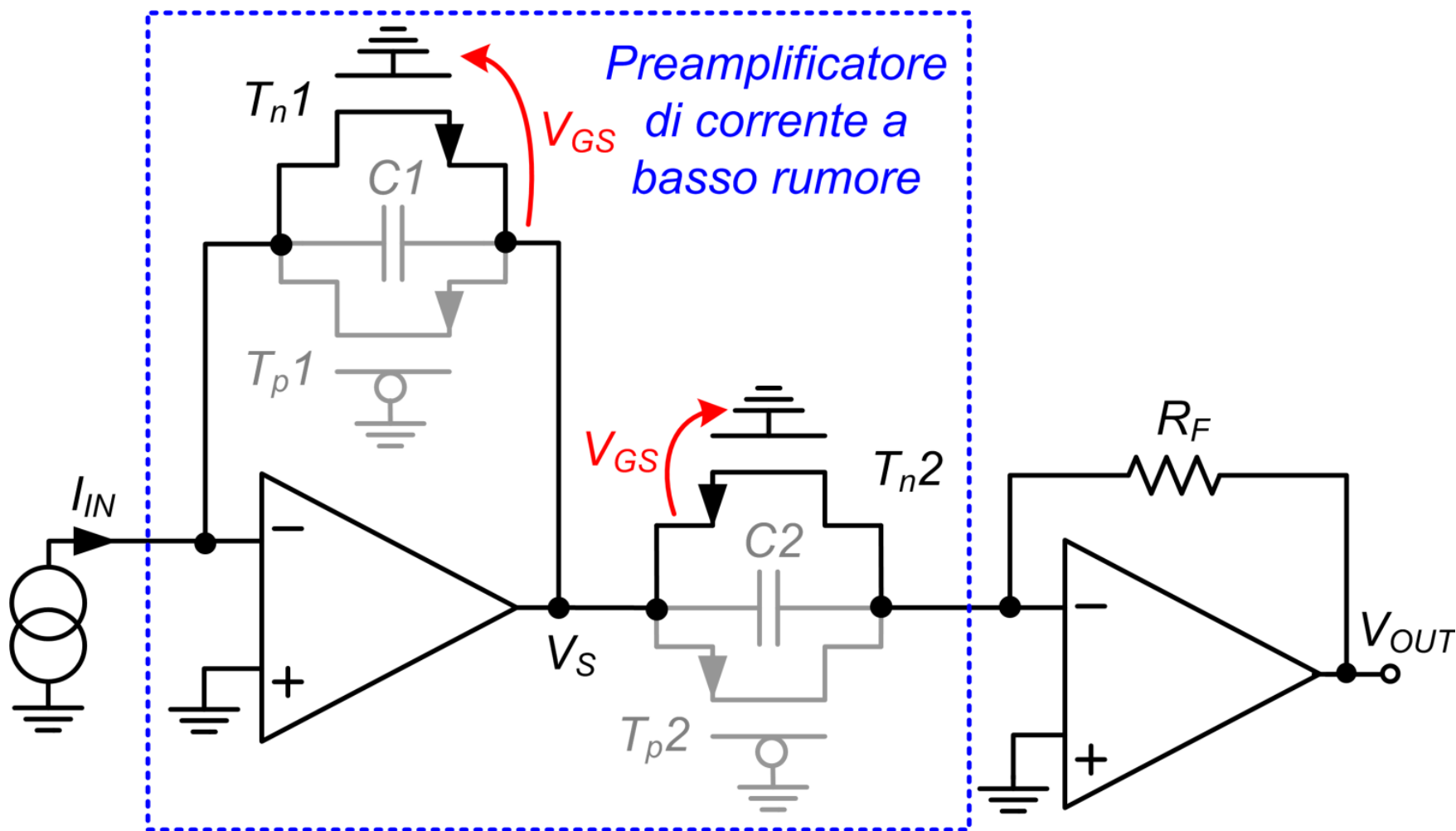


$$\overline{i_{TOT}^2} \cong 2qI_{IN} + \overline{e_n^2} \cdot (2\pi f)^2 \cdot (C1 + C_{IN})^2$$



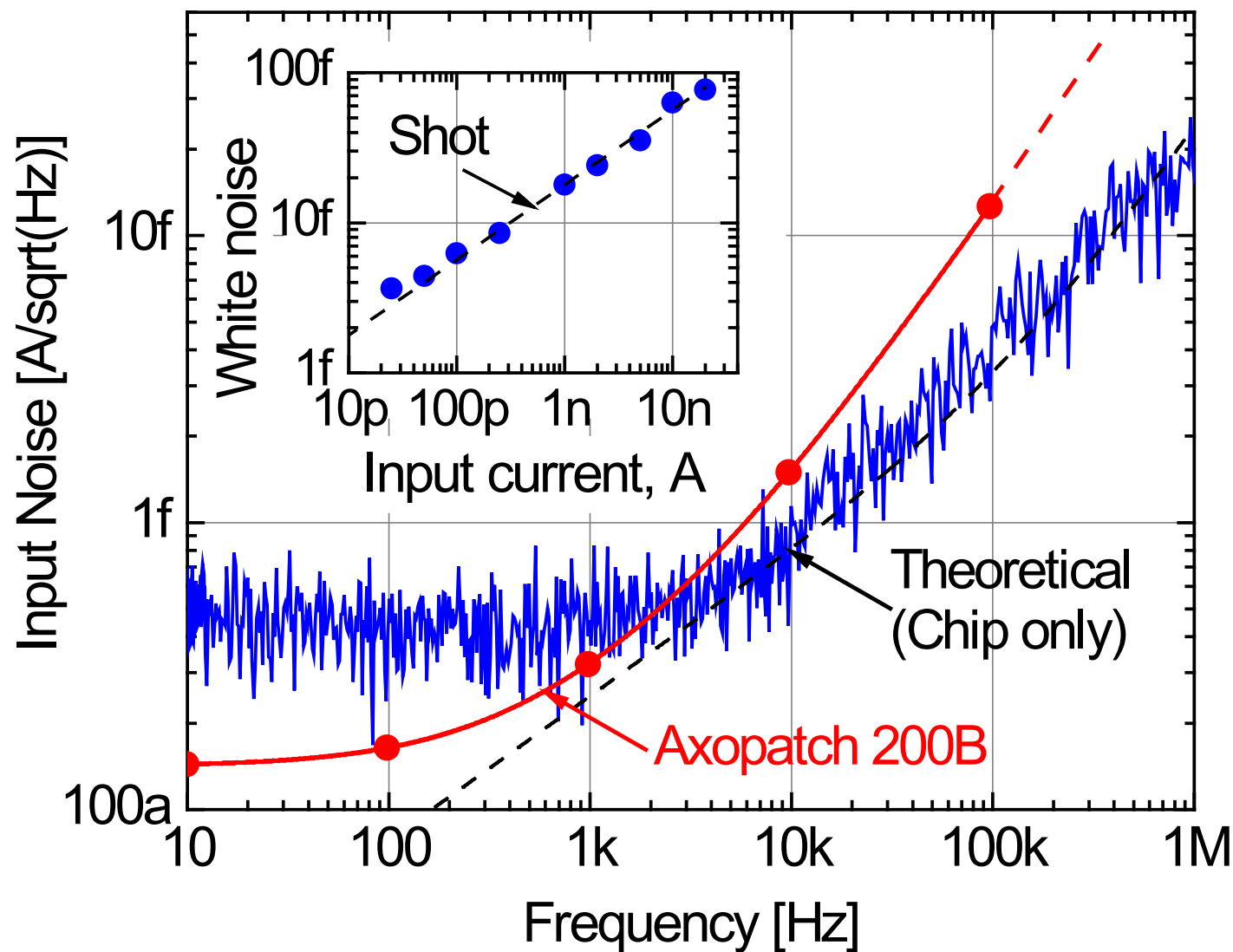
# Low-Noise Wide-Bandwidth Amplifier

Equivalent to a larger feedback resistor  $R_F \cdot G$  (but wider BW):





# Example of Experimental Performance







REVIEW OF SCIENTIFIC INSTRUMENTS 78, 094703 (2007)

## Wide bandwidth transimpedance amplifier for extremely high sensitivity continuous measurements

Giorgio Ferrari<sup>a)</sup> and Marco Sampietro

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

1

## Noise Limits of CMOS Current Interfaces for Biosensors: A Review

Marco Crescentini, *Member, IEEE*, Marco Bennati, Marco Carminati, *Member, IEEE*, and Marco Tartagni, *Member, IEEE*

## Ultra-low-noise CMOS current preamplifier from DC to 1 MHz

G. Ferrari, M. Farina, F. Guagliardo, M. Carminati and M. Sampietro